

LA's BeST @USC

Single-Cell RNA Sequencing Project

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Southern California

The Central Dogma

DNA



transcription



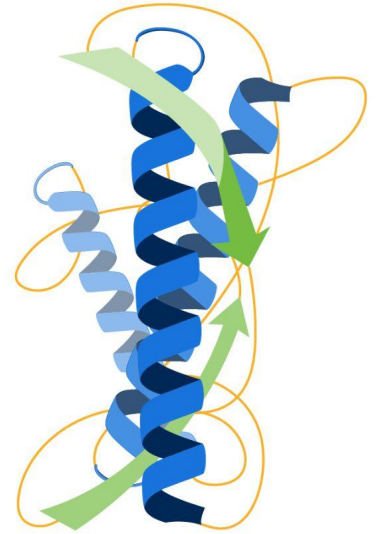
RNA



translation



Protein



Cell Types

44.5M Cells

ALL CELLS

Blood

Kidney

Bone

Liver

Brain

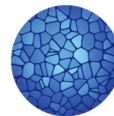
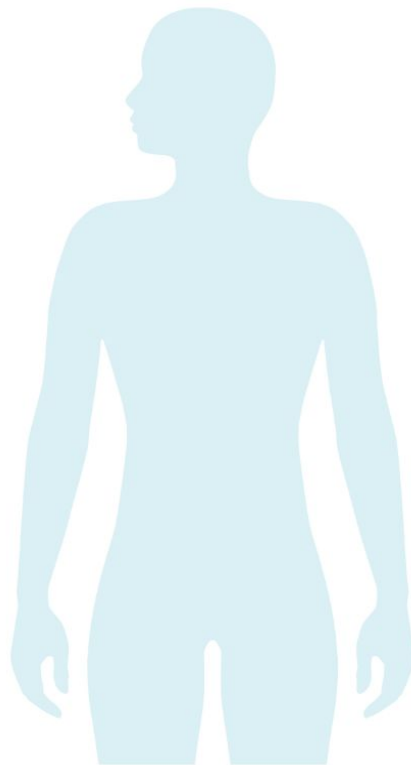
Lung

Pancreas

Heart

Immune System

Skin



HUMAN
CELL
ATLAS

Why Single Cell?

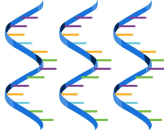
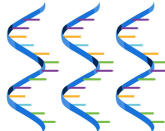


Bulk



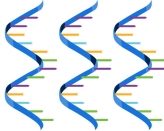
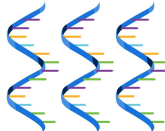
Single Cell

Gene Expression

	 heart cell	 lung cell	 blood cell
gene A			
gene B			
gene C			
⋮	⋮	⋮	⋮

The Human Genome Project
estimated that we have
20,000 - 25,000 genes

Gene Expression

	 heart cell	 lung cell	 blood cell
gene A			
gene B			
gene C			
⋮	⋮	⋮	⋮

Single-Cell RNA Sequencing

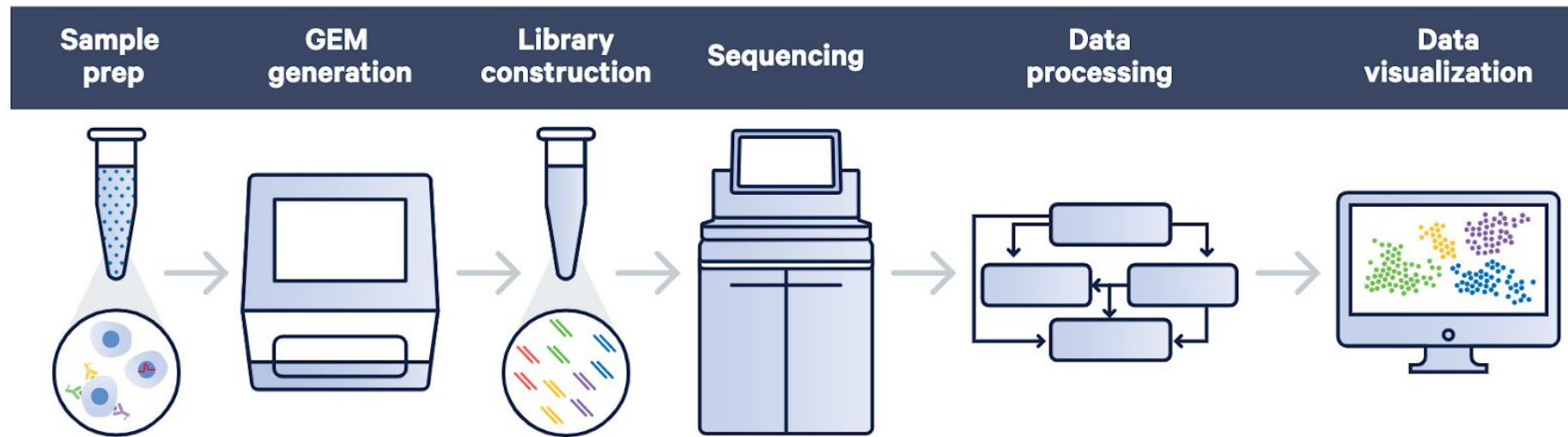
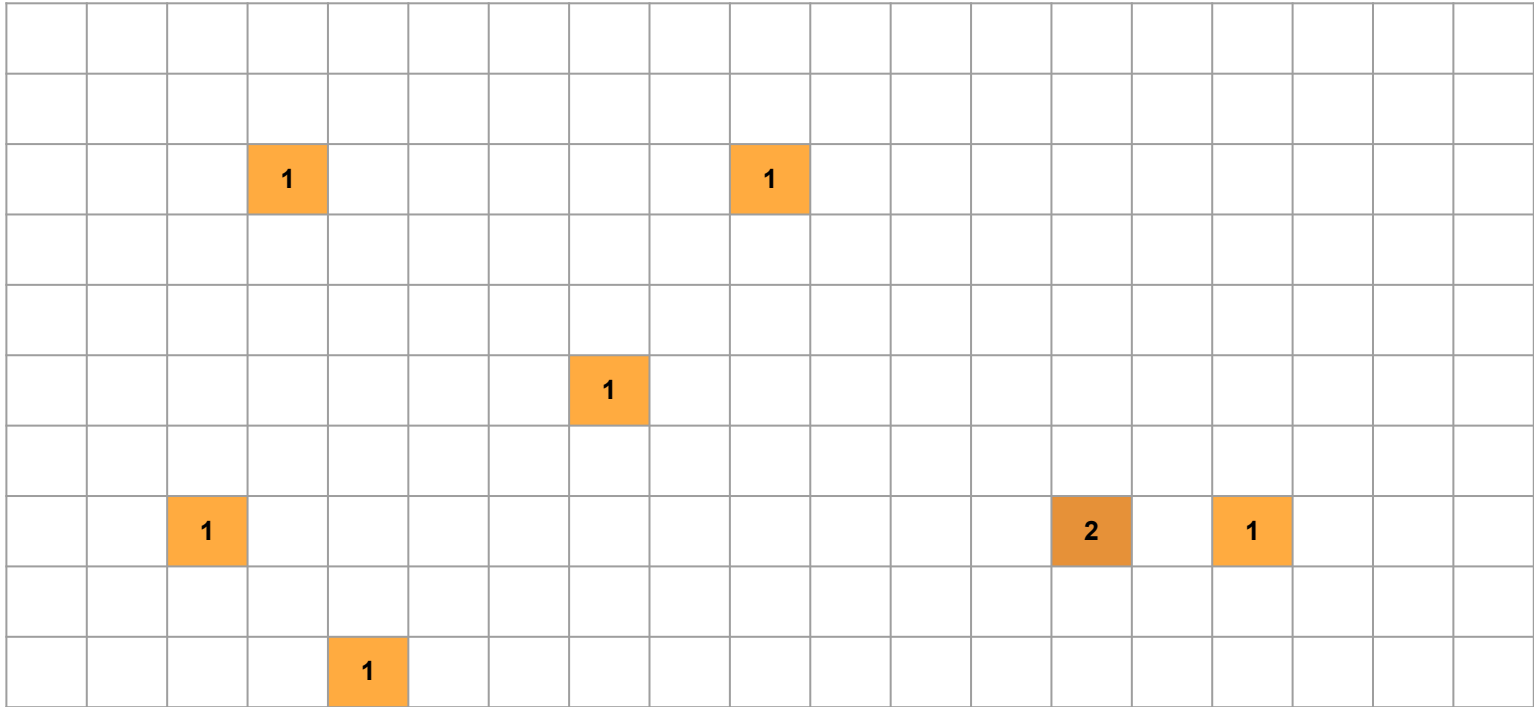
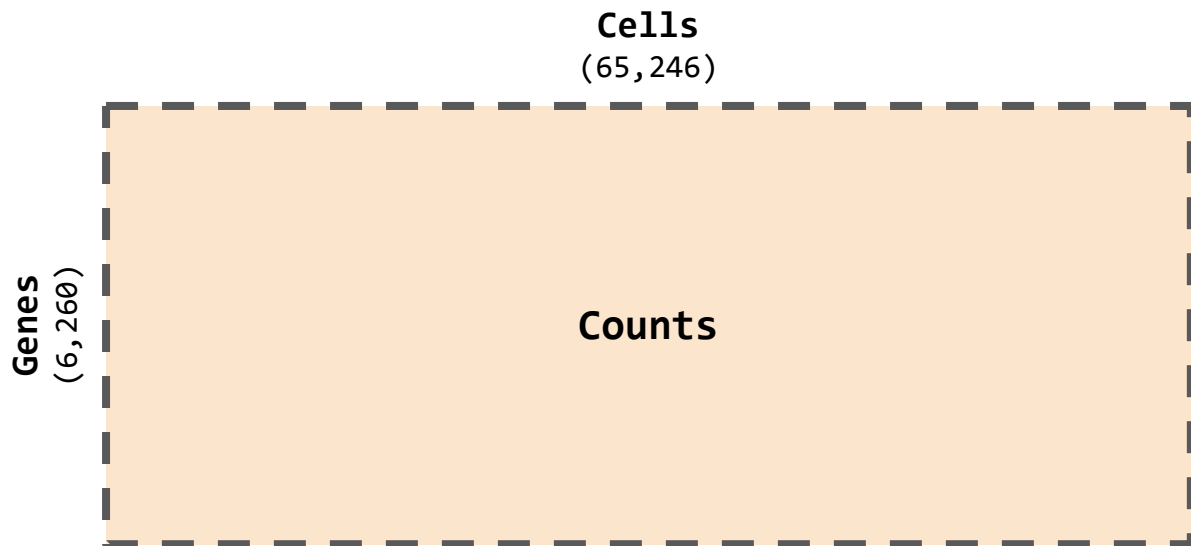


Figure 2. Efficient and streamlined workflow for multiomic profiling of biological systems. Start with a single cell suspension of unlabeled cells, oligo-conjugated antibody-labeled cells, or cells expressing compatible CRISPR guides. After GEM generation, separate libraries can be constructed from a single sample, including gene expression and cell surface protein or CRISPR guide libraries, generating multiple readouts that can be linked back to the same single cell. Process data with Cell Ranger, and visualize sample heterogeneity with Loupe Browser, our fully integrated and easy-to-use analysis and visualization software.



scRNAseq Data



Article

Progressive immune dysfunction with advancing disease stage in renal cell carcinoma

David A. Braun,^{1,2,3,17} Kelly Street,^{4,5,17} Kelly P. Burke,^{1,2,6} David L. Cookmeyer,^{2,6} Thomas Denize,^{2,7} Christina B. Pedersen,^{8,9} Satyen H. Gohil,^{1,2,3,10} Nicholas Schindler,¹ Lucas Pomerance,^{1,2} Laure Hirsch,^{1,2} Ziad Bakouny,¹ Yue Hou,^{1,11} Juliet Forman,^{1,3,11} Teddy Huang,^{1,11} Shuqiang Li,^{1,3,11} Ang Cui,^{3,12} Derin B. Keskin,^{1,3,11} John Steinharter,¹ Gabrielle Bouchard,¹ Maxine Sun,¹ Erica M. Pimenta,^{1,2} Wenxin Xu,^{1,2} Kathleen M. Mahoney,^{1,2,13} Bradley A. McGregor,^{1,2} Michelle S. Hirsch,^{2,7} Steven L. Chang,^{2,14} Kenneth J. Livak,^{1,11} David F. McDermott,^{2,13} Sachet A. Shukla,^{3,11} Lars R. Olsen,^{8,9} Sabina Signoretti,^{2,7,15} Arlene H. Sharpe,^{2,6,7,16} Rafael A. Irizarry,^{4,5} Toni K. Choueiri,^{1,2,18} and Catherine J. Wu^{1,2,3,18,19,*}

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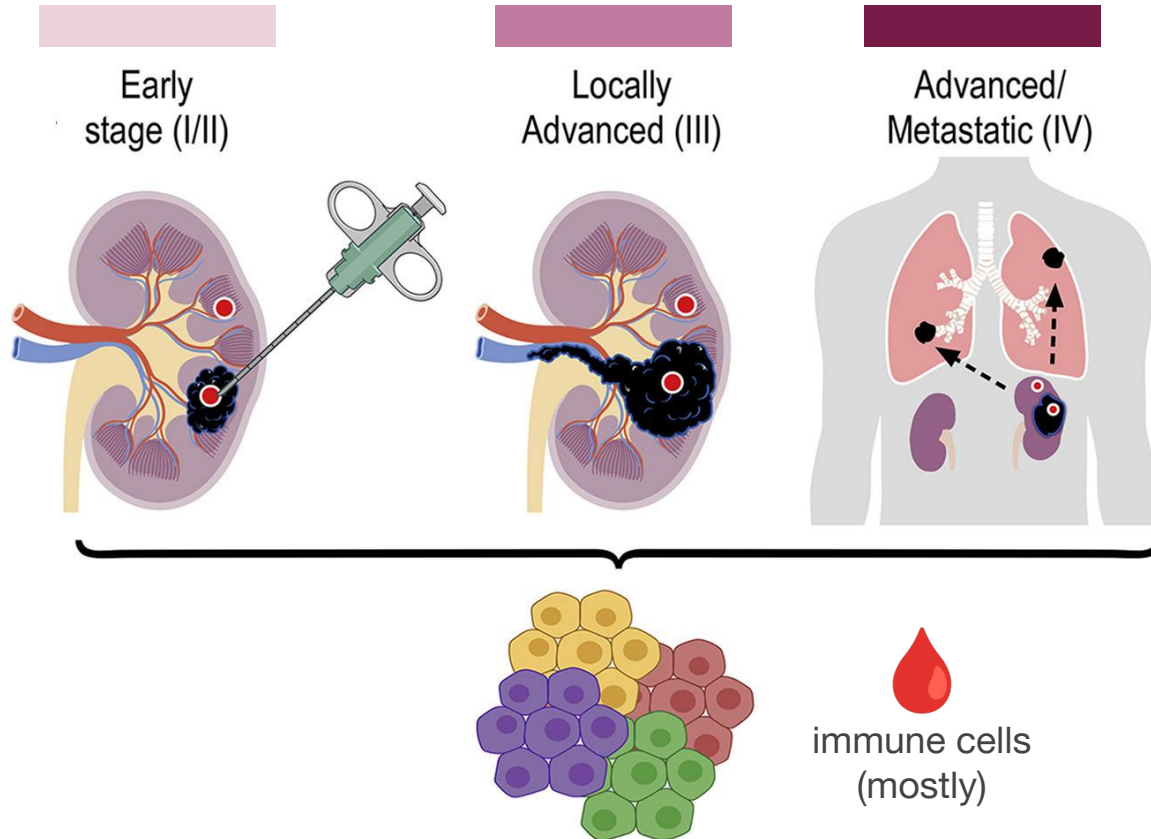
⁵Department of Biostatistics, Harvard T.H. Chan School of Public Health, Boston, MA 02215, USA

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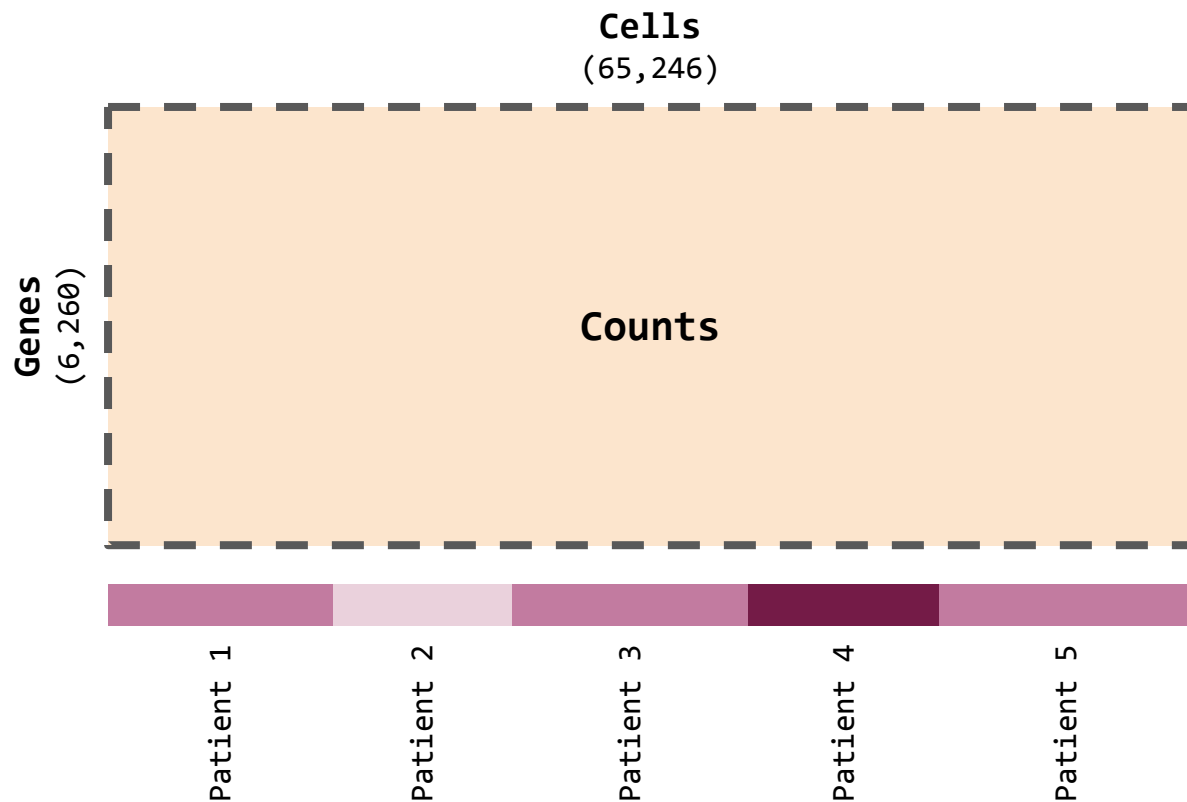
⁷Department of Pathology, Brigham and Women's Hospital, Boston, MA 02115, USA

⁸Section for Bioinformatics, Department of Health Technology, Technical University of Denmark, Kongens Lyngby, Denmark

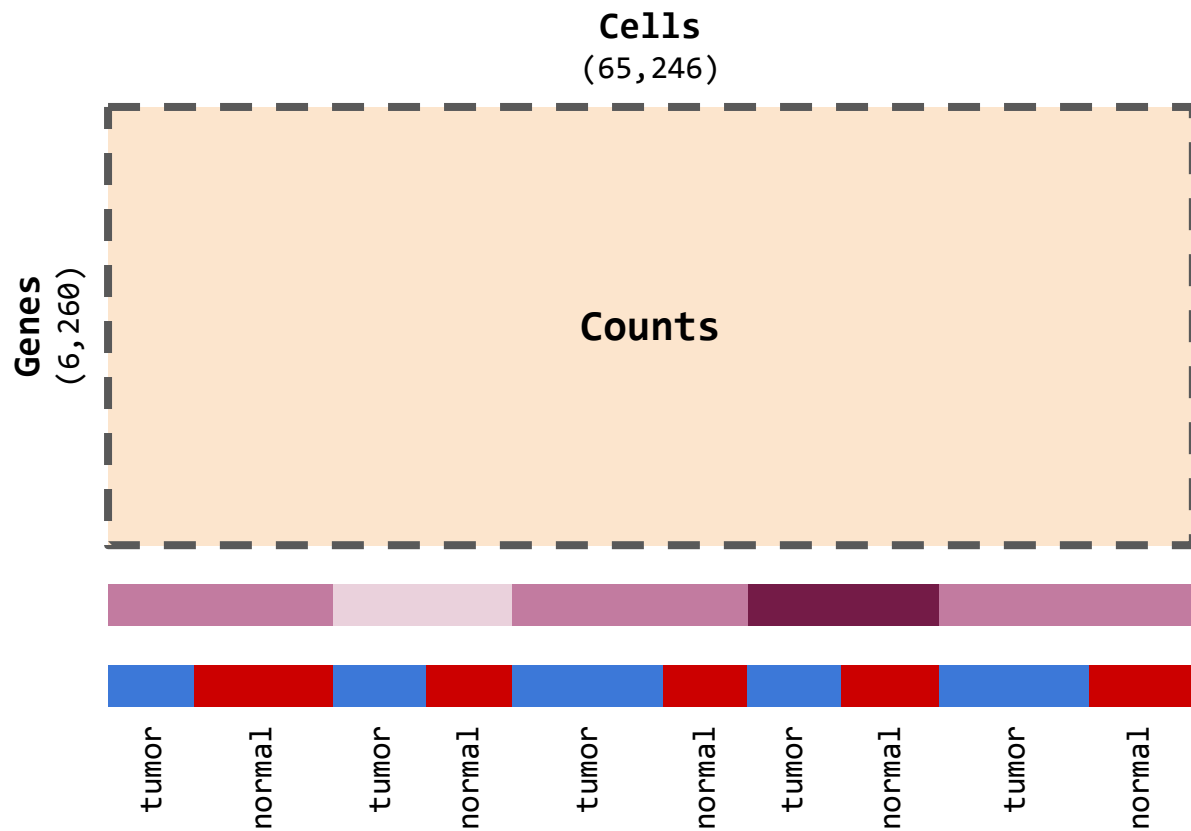
Renal Cell Carcinoma



Gene Expression



Gene Expression

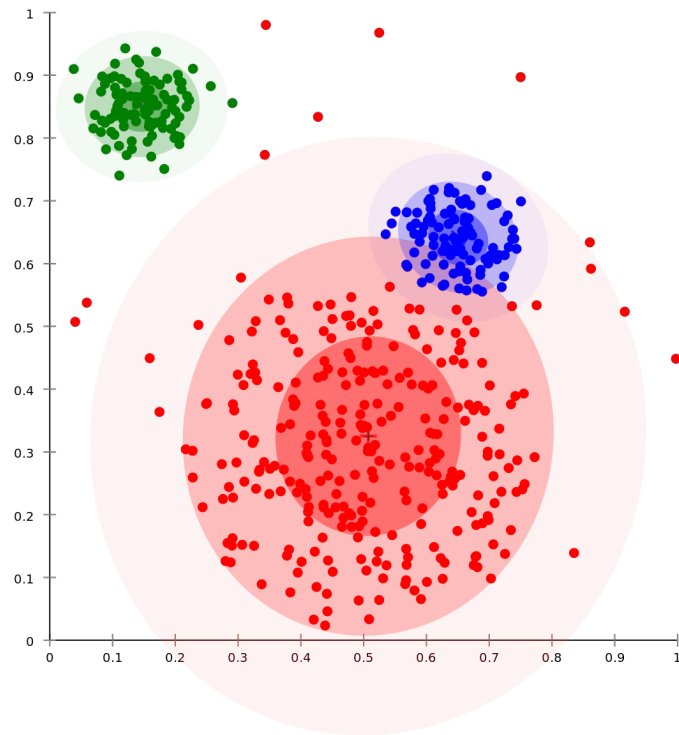


Question 1:

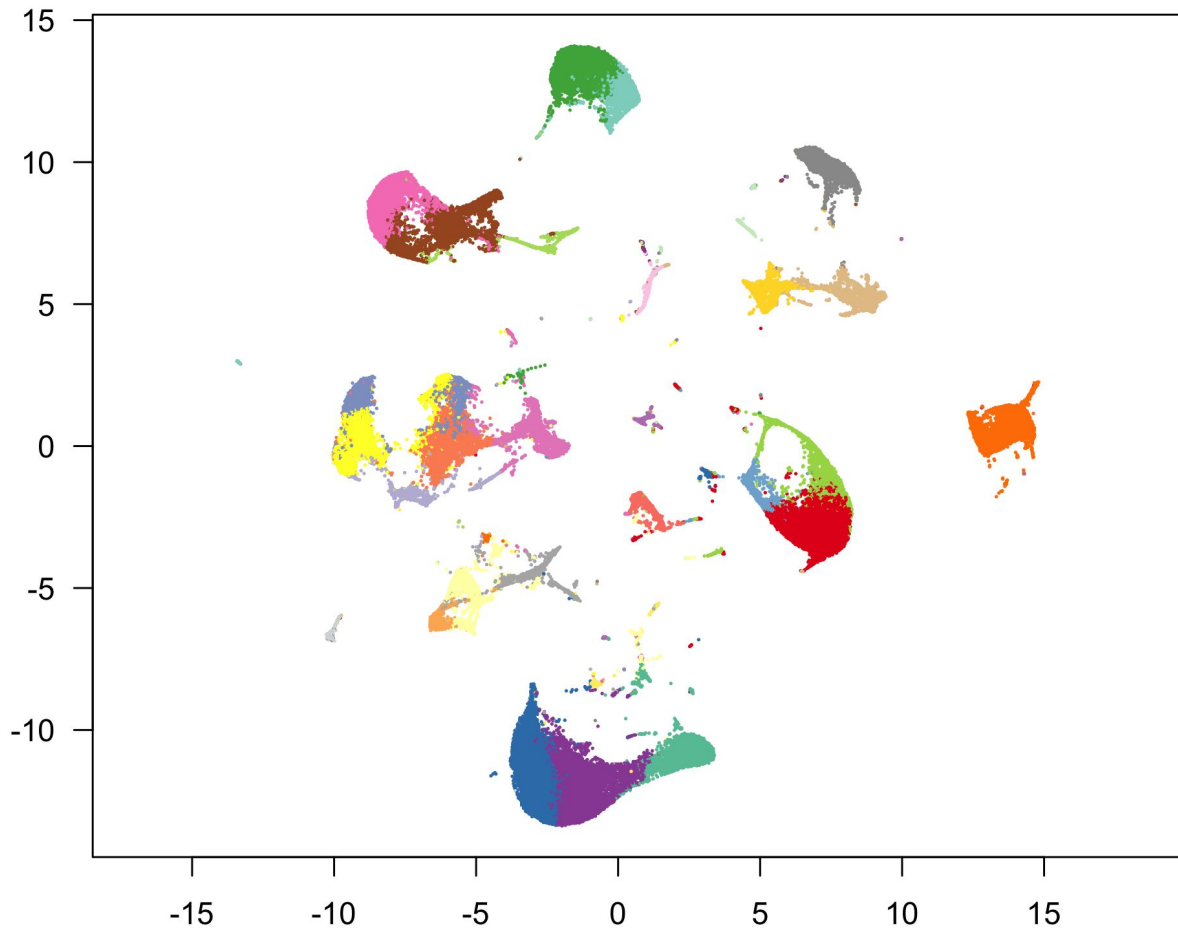
What **cell types** are present?

Clustering

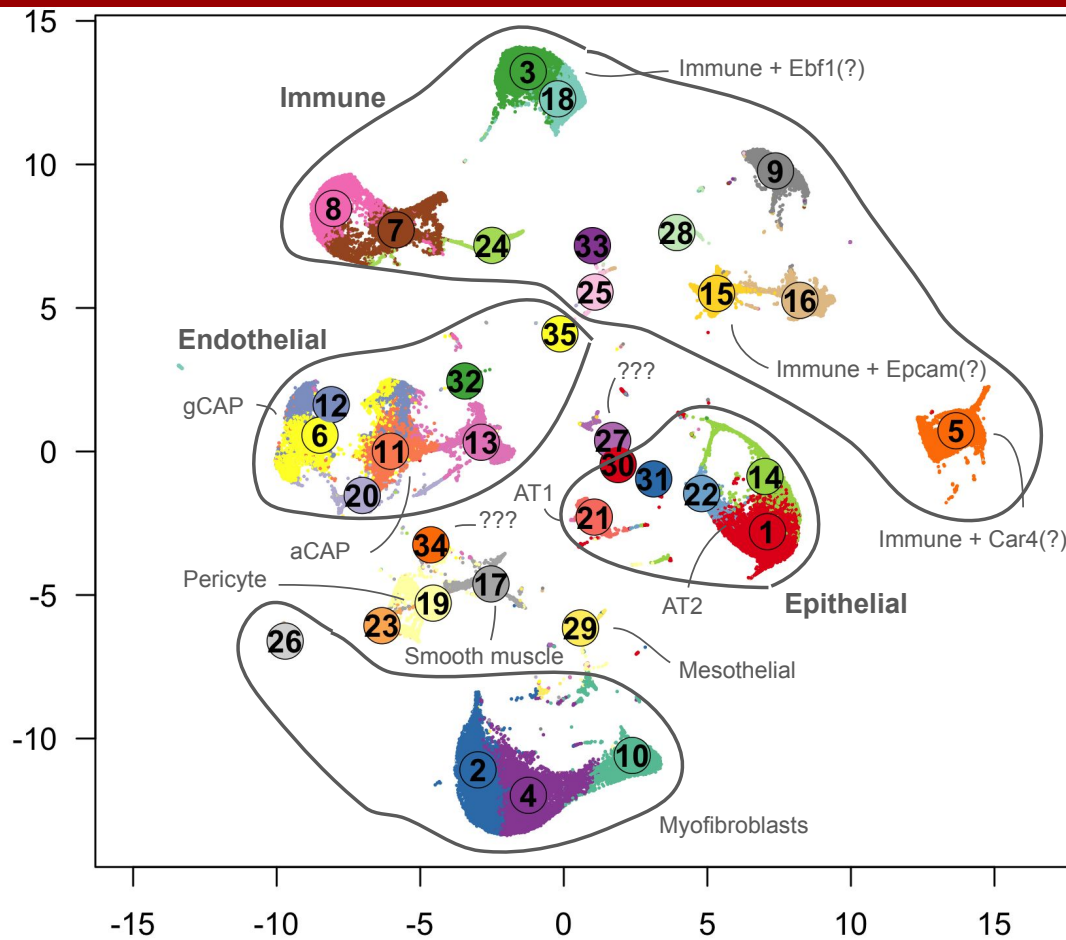
Identify groups of cells with **similar** expression profiles



Clustering



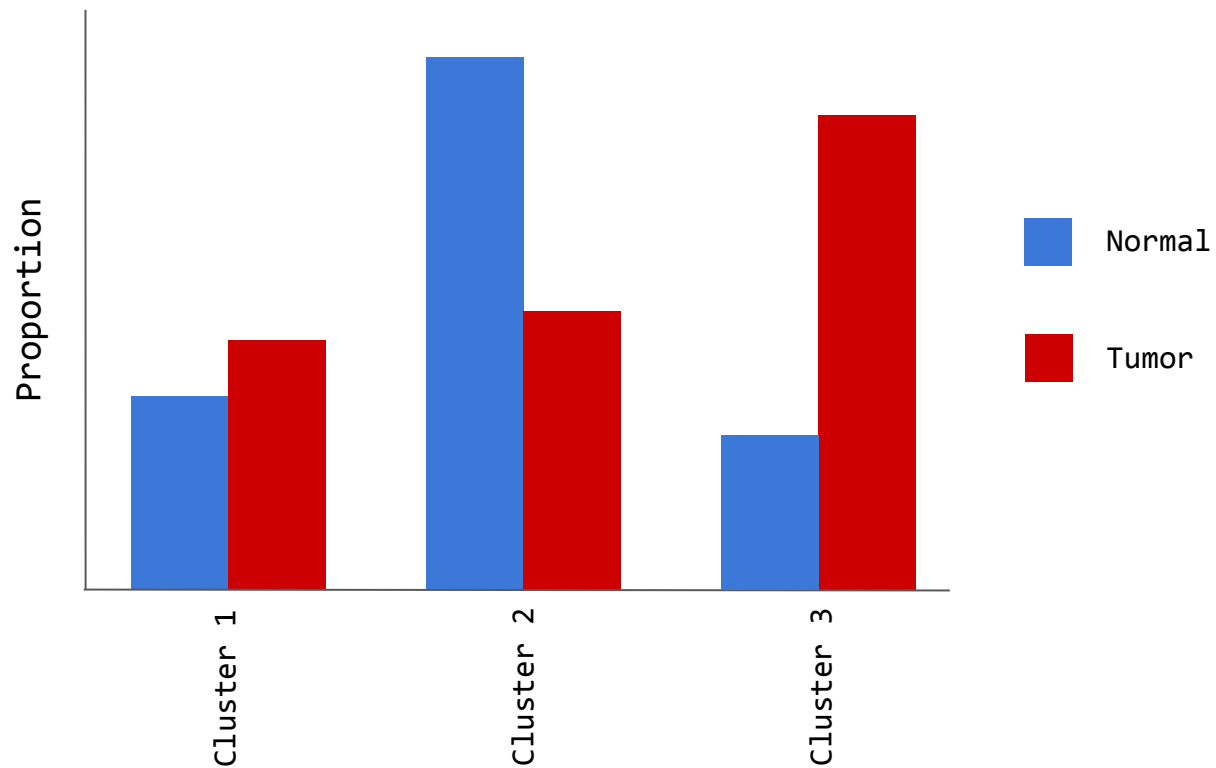
Clustering



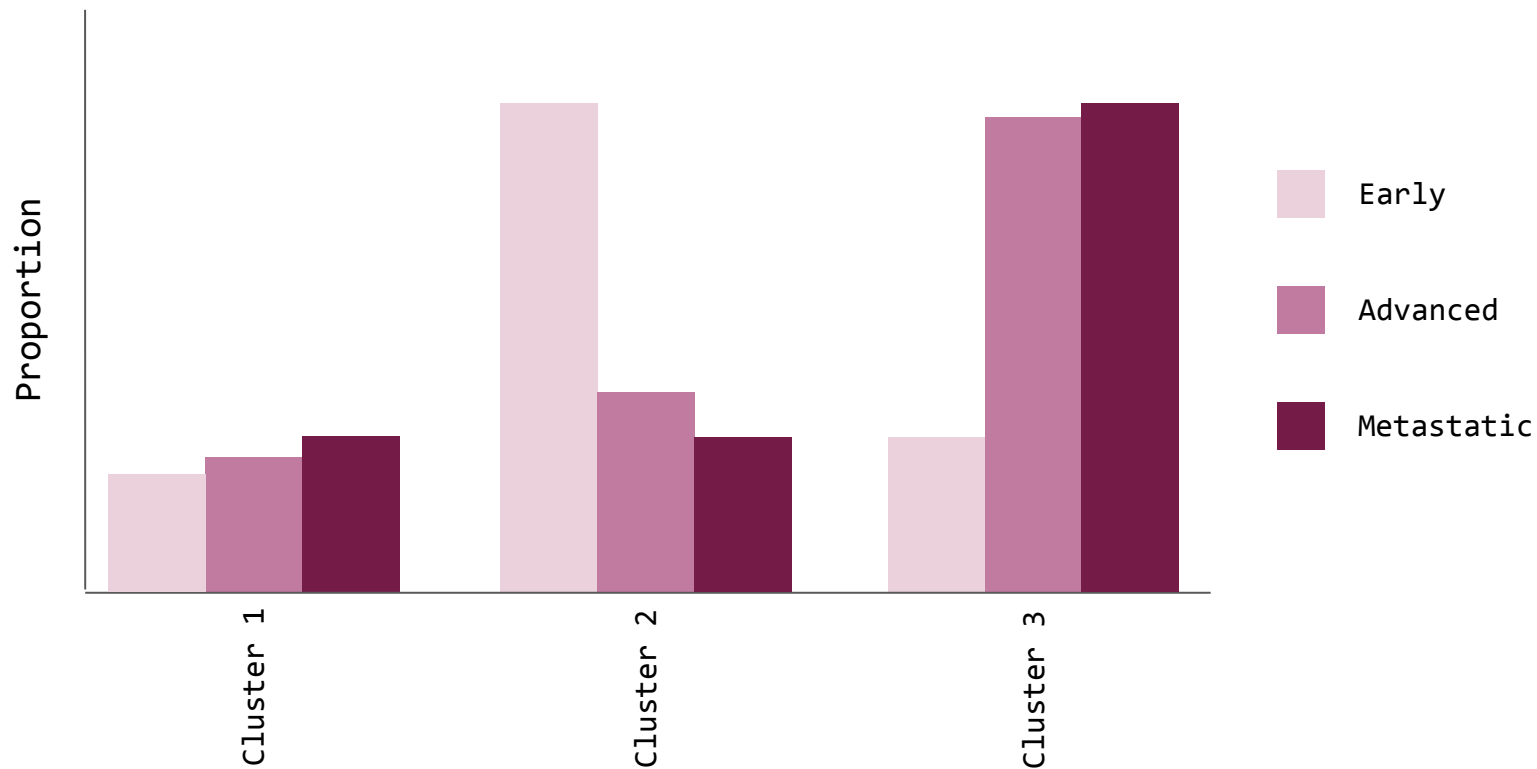
Question 2:

Which cell types are
associated with disease?

Differential Abundance



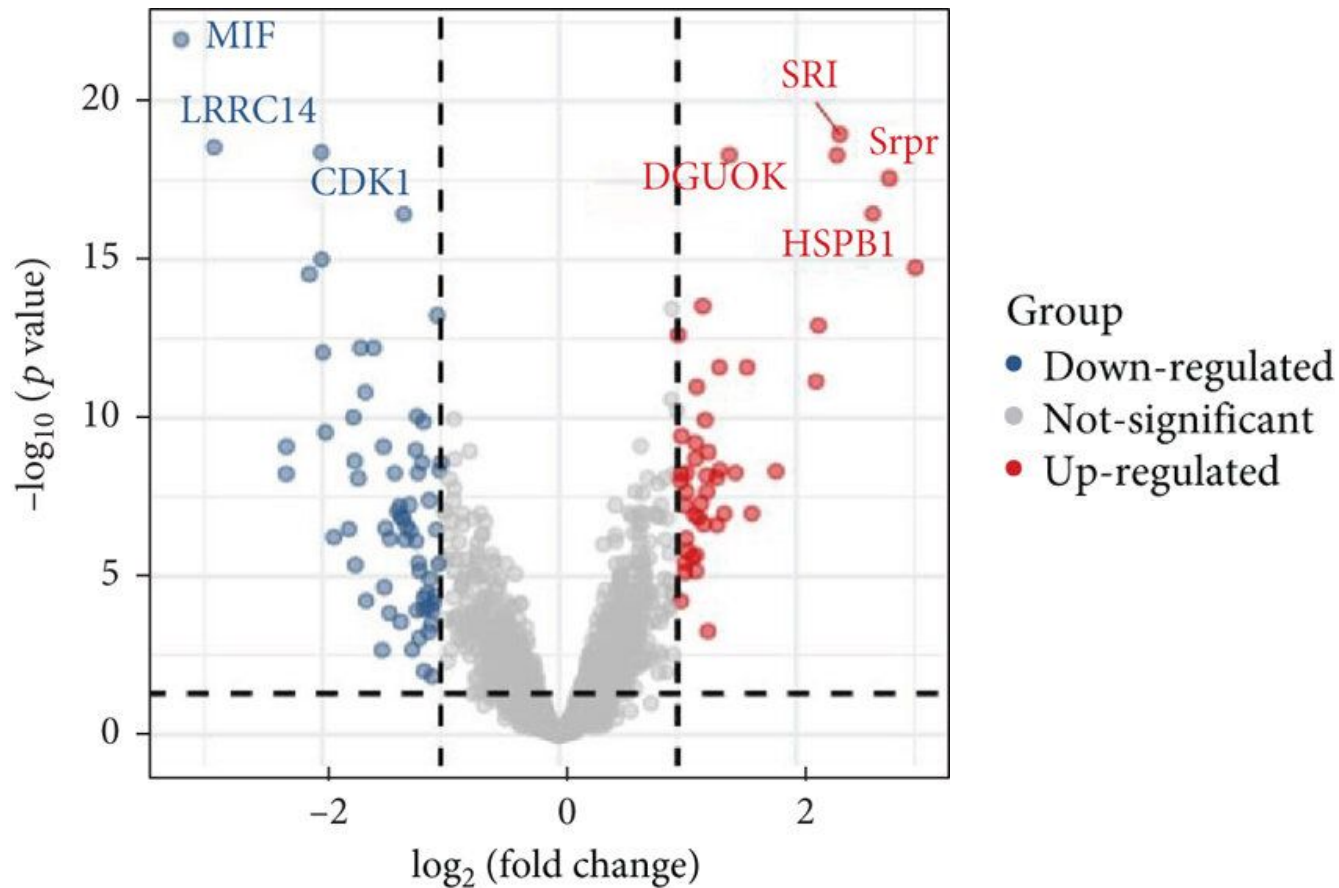
Differential Abundance



Question 3:

Which genes **change their expression** with disease?

Differential Expression



Differential Expression

